

Stress-Testing Prospective Multimodal Dataset Utility Prediction

Anonymous authors
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Abstract

Can candidate multimodal data be screened before paying the cost of training on them? We study this question under an explicit information boundary: a selector may inspect candidate data, target-training anchors, and model metadata, but not candidate transfer outcomes or target-validation labels. Our evaluation uses common text and speech representations, three sentiment and emotion corpora, five seeds, three separately salted repartition families, and four downstream learners. Under a fixed logistic-regression recipe, centroid distance identifies the oracle source in 44 of 45 repeated decisions, although these decisions cover only three independent target corpora. The result does not extend to choosing a bucket within that source. A positive-utility classifier improves mean utility by 0.00105 over a short-curve baseline, with a target-cluster 95% interval of $[-0.00380, 0.00562]$ and an exact p -value of 0.50. Independent calibration finds no nonvacuous rejection threshold. Matched GaussianNB, neural-MLP, and learned gated-fusion panels further show that utility prevalence, rankings, and source routes change with the downstream learner. The evidence supports a narrow conclusion: source compatibility can be predictable for a fixed recipe, whereas within-source utility estimates and confidence thresholds do not transfer reliably across partitions or learners.

1 Introduction

Multimodal training sets are often assembled from heterogeneous corpora. More data is not automatically better: an added source can raise training cost, shift the class distribution, or reduce performance on the target task. The useful decision is therefore prospective. Given many candidate sources, which ones should be evaluated at all? A score derived from their eventual transfer outcomes arrives too late to save computation.

Generic notions of data quality do not settle this decision. Clean labels and well-aligned modalities say little about compatibility with a particular target. Compatibility may also change with the downstream learner. We formalize this dependence as $\mathcal{U}(\mathcal{S}; \mathcal{T}, M, R)$, where source $\mathcal{S}$, target $\mathcal{T}$, model M , and training recipe R jointly determine utility. Calling a dataset “high quality” without fixing the remaining arguments obscures the quantity that selection is meant to predict.

This paper examines the limits of prospective multimodal data selection. We separate corpus-level source routing from within-source bucket selection, since a representation may distinguish domains while failing to order nearby subsets. We also evaluate abstention separately from ranking. A selector that rejects every candidate avoids harm, but it does not support data acquisition. Learned scores are tested on held-out target corpora or sealed repartitions, and utility labels are materialized only after predictions have been frozen.

The paper makes three contributions:

- We define multimodal dataset utility with a strict information boundary and provide a deterministic five-seed protocol built on common text and audio representations.

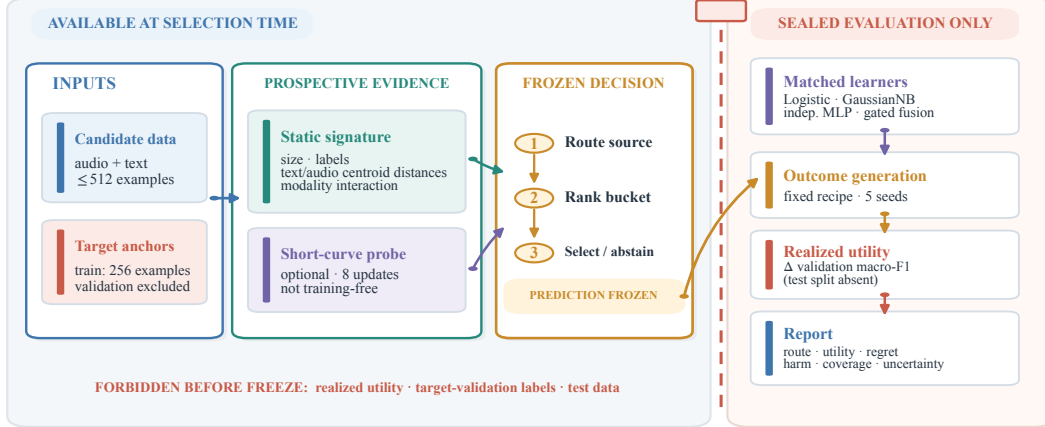


Figure 1: Information boundary and evaluation flow. Candidate data and target-training anchors produce static signatures and, when enabled, an eight-update probe. The probe is not training-free. Routes, bucket rankings, and abstention decisions are frozen before candidate utilities or target-validation labels are revealed. Full downstream training produces only the sealed evaluation outcomes.

- We isolate a granularity gap. For logistic regression, source routing is stable across repeated partitions (44/45 decisions over three independent target corpora), while bucket ranking and calibrated abstention are not.
- We measure target-level uncertainty and repeat the same candidate comparisons with GaussianNB, neural MLP, and learned gated fusion. The matched panels reveal substantial learner dependence that candidate-level averages would otherwise conceal.

2 Related Work

Transferability estimation. LEEP, LogME, and TransRate estimate the transferability of a pretrained model or representation without exhaustive fine-tuning (Nguyen et al., 2020; You et al., 2021; Huang et al., 2022). These methods rank source models or feature spaces. We instead hold common representations fixed and rank candidate training data by the change it induces in a specified downstream learner. This distinction matters because data utility can change with the learner even when the representation and target are unchanged.

Data valuation and task-aware selection. Data Shapley assigns value through marginal downstream utility, whereas datamodels learn a mapping from training subsets to model predictions (Ghorbani & Zou, 2019; Ilyas et al., 2022). DsDm uses this model-aware view for language-model dataset selection, and LESS uses optimizer-aware gradient similarity for targeted instruction tuning (Engstrom et al., 2024; Xia et al., 2024). Together, these approaches show that usefulness can depend on the target and learner, but their scores require subset-training supervision, warmup training, or gradients. Our static signatures obey a stricter pre-candidate-training boundary; short probes are reported separately rather than described as training-free.

Multimodal curation. DataComp standardizes image-text filtering under fixed compute, while JEST selects jointly learnable multimodal batches using learner and reference losses (Gadre et al., 2023; Evans et al., 2024). Recent preprints further combine intrinsic quality, relevance, diversity, validation feedback, capability-specific raters, or one-step proxy updates (Cheng et al., 2026; Sahi et al., 2026; Jing et al., 2026). They primarily optimize large-scale vision-language pretraining or multimodal instruction tuning. Our setting instead asks

whether whole audio–text sources and their internal buckets can be screened before running a fixed target augmentation recipe.

Training-free supervision-data generalization. DataProphet is the closest precedent. It predicts influence rankings for vision–language supervision datasets from perplexity, similarity, and diversity, without candidate-specific training (Qi et al., 2026). We do not claim the first training-free multimodal selector. Instead, we test how far this premise extends when selection moves from corpora to buckets and when the downstream learner changes. The evaluation adds sealed natural repartitions, target-cluster uncertainty, and independently calibrated abstention.

3 Problem Formulation

Let $D_{\mathcal{T}}^{\text{tr}}$ and $D_{\mathcal{T}}^{\text{val}}$ denote target training and validation data, and let $\mathcal{S}_i$ be a candidate source block. For a fixed model and recipe (M, R) , we define realized utility under seed s as

$$u_{i,s} = q(M_R(D_{\mathcal{T}}^{\text{tr}} \cup \mathcal{S}_i; s), D_{\mathcal{T}}^{\text{val}}) - q(M_R(D_{\mathcal{T}}^{\text{tr}}; s), D_{\mathcal{T}}^{\text{val}}), \quad (1)$$

where q is macro-F1. Ground-truth utility is $\mathcal{U}_i = |\mathcal{K}|^{-1} \sum_{s \in \mathcal{K}} u_{i,s}$ over five locked seeds. Positive utility means that adding the candidate improves validation performance on average.

The selector may use candidate data, target-training anchors, model metadata, and features derived from them. It may not use candidate-specific $u_{i,s}$, target validation labels, or target test data. Given scores $\hat{u}_i$, forced selection chooses $\arg \max_i \hat{u}_i$. We report selected utility and Top-1 regret

$$\text{Regret} = \max_i \mathcal{U}_i - \mathcal{U}_i. \quad (2)$$

A selective rule may abstain, in which case its realized utility is zero. We report its coverage and accepted harm, defined by $\mathcal{U}_i < 0$.

4 Evaluation Framework

4.1 Data and common representations

The multimodal panel comprises CMU-MOSI (Zadeh et al., 2016), MELD (Poria et al., 2019), and CH-SIMS (Yu et al., 2020). Only audited training and validation splits enter the study. We map sentiment labels to negative, neutral, and positive classes. Text is represented by the frozen 384-dimensional paraphrase-multilingual-MiniLM-L12-v2 sentence transformer (Reimers & Gurevych, 2019; Wang et al., 2020); audio is encoded by the frozen 384-dimensional whisper-tiny encoder (Radford et al., 2022). Model revisions are fixed in the released manifests, and candidate and target IDs are aligned before transfer experiments. Target anchors contain 256 examples and candidate blocks contain at most 512 examples.

The development partition hashes video or dialogue identifiers and therefore keeps groups intact. The three confirmation families instead hash salted sample identifiers. They are deterministic, but they do not preserve groups. Source examples can move between salts and reappear in another family, so salts and families are repeated measurements rather than independent datasets.

For each target, 256 training anchors define the baseline. Candidate data are appended with total source weight capped at the anchor weight. Separate text and audio heads produce three-class probabilities, followed by equal-weight fusion. Utility is the change in validation macro-F1 from the anchor-only baseline, averaged over five fixed seeds.

4.2 Prospective signatures and hierarchical decisions

The static signature records candidate size, label statistics, target–source centroid distances for text and audio, and cross-modal interaction descriptors. An eight-update probe adds learning-curve features from support and query splits of the target-training data. The probe

never accesses target-validation data or a candidate’s realized utility, but it is not training-free.

Selection proceeds in two stages. Source routing chooses the corpus with the smallest mean text/audio centroid distance across its eight buckets. Within that source, Curve-19 ranks buckets by Ridge regression on the 19-dimensional probe signature. A balanced logistic classifier provides a second score by predicting $\mathcal{U}_i > 0$. Both selectors are fit in leave-one-target-out folds. Standardization within a target-source stratum uses candidate features but no utilities.

4.3 Sealed confirmation and uncertainty

Each of the three separately salted families contains 240 candidates: three targets, two eligible sources, five salts, and eight buckets. The families are protocol-separated but not sample-disjoint. Prediction artifacts are written before utility generation. Rejection thresholds are chosen on a dedicated calibration family and evaluated once on the final family.

Partition decisions share corpora and are not independent replicates. For primary uncertainty estimates, effects are first averaged within targets and then entered into a 10,000-draw hierarchical bootstrap over targets, batches, and salts. We also enumerate all 2^3 target-level sign flips. Three target clusters cannot support exact significance at 0.05, and salt-level tests would overstate the effective sample size.

4.4 Downstream learners

The primary recipe uses class-balanced logistic-regression heads for text and audio. Matched panels replace both heads with GaussianNB or independent two-layer MLPs while holding candidates, weights, representations, seeds, and the validation metric fixed. Each MLP has a 64-unit ReLU hidden layer and receives 80 deterministic, full-batch AdamW updates. A fourth panel learns fusion jointly: modality-specific 384-to-64 ReLU branches feed a two-way softmax gate and a shared three-class head, again trained for 80 fixed updates. GaussianNB, independent-MLP, and gated-fusion anchor macro-F1 ranges are 0.442–0.532, 0.437–0.540, and 0.423–0.543, respectively, ruling out completely degenerate comparisons.

4.5 Compute accounting

Table 1 separates exact training counts from machine-dependent timings. The three sealed logistic families and the matched GaussianNB and MLP panels contain 6,000 candidate-seed evaluations, requiring 12,150 fitted modality heads after shared anchor baselines are included. The gated-fusion panel adds 1,200 candidate-seed evaluations and 1,215 jointly trained models, or 97,200 full-batch updates. In a deterministic rerun of one 240-candidate panel, Curve-19 screening takes 14.46 seconds, compared with 17.88 seconds for full logistic utility generation. The 19.1% wall-time reduction is modest because both methods are inexpensive in this classical setting. All 1,200 stored utility rows are reproduced exactly.

For the motivating 100-to-10 decision, exhaustive five-seed evaluation with two heads requires 1,000 candidate head fits. Evaluating ten selected candidates requires 100, avoiding 90% of candidate fits, or 89.1% after a shared ten-fit baseline is included. This calculation is conditional; it is neither an observed speedup nor evidence that the selected ten are optimal. End-to-end neural savings remain unmeasured.

5 Results

5.1 Source routing is easier than bucket selection

The hierarchy localizes the main error. Routing followed by the original bucket score lowers regret from 0.04278 to 0.02730, yet mean utility remains negative (Table 2). An oracle restricted to the routed source attains positive utility on all three targets with regret 0.00060. Under the logistic recipe, useful buckets are present in the routed source; the selector fails to identify them reliably.

Table 1: Audited computation. Counts are exact; wall times are local CPU measurements on the same 240-candidate panel and are not estimates of neural-model cost. Each fused evaluation fits separate text and audio heads.

Stage	Blocks	Primitive computation	Wall time (s)
Curve-19 screening	240	9,600 fused updates	14.46
One logistic utility panel	240	1,200 seeded evaluations	17.88
Three logistic panels	720	7,290 head fits	—
Matched GaussianNB panel	240	2,430 head fits	—
Matched neural MLP panel	240	2,430 head fits	83.55
Complete reported audit	1,200	12,150 head fits	—

Table 2: Hierarchical error decomposition on the initial three-target panel. Oracle diagnostics are not deployable. The small routed-source oracle regret isolates within-source bucket discrimination as the logistic-model bottleneck.

Decision rule	Mean utility $\uparrow$	Regret $\downarrow$	Positive targets
Flat original-12	-0.02800	0.04278	0/3
Hierarchical original-12	-0.01252	0.02730	1/3
Routed-source oracle bucket	+0.01417	0.00060	3/3

5.2 Apparent gains reverse across repartitions

On the first sealed family, the positive-utility classifier lowers regret from 0.01780 to 0.01211 relative to Curve-19 and selects a beneficial bucket in 10 of 15 decisions. The next two families reverse the gain, with utility differences of -0.00153 and -0.00102 (Figure 2). Across all 45 decisions, the mean difference is 0.001047; its 95% hierarchical interval is $[-0.003804, 0.005616]$, and the exact target-level test gives $p = 0.50$.

A post-hoc baseline audit reaches the same cautionary conclusion. Uniform bucket selection has expected utility 0.00077 and regret 0.01810, compared with 0.00178/0.01727 for the classifier and 0.00073/0.01832 for Curve-19 (Table 3). Curve-19 does not outperform random selection on these aggregate metrics. Because the random baseline was added after the primary outcomes were inspected, the classifier’s margin over it remains descriptive.

5.3 Classifier confidence does not support safe abstention

The calibration family yields no eligible probability threshold. Between 0.50 and 0.99, accepted harm ranges from 50% to 75%, and realized utility is negative at every grid point (Figure 3). The locked fallback threshold of 1.0 rejects every final decision. It prevents accepted harm only by reducing both coverage and realized utility to zero.

5.4 Utility rankings depend on the downstream learner

Matched logistic-regression and GaussianNB utilities agree moderately in rank, but not in sign (Figure 4). Beneficial candidates account for 28.75% of the logistic panel and only 3.75% of the GaussianNB panel. On the same family, route accuracy falls from 14/15 to 10/15. Both frozen bucket selectors choose harmful GaussianNB candidates in all 15 decisions (Table 3).

The MLP panel preserves some coarse structure but little within-source agreement. The distance route matches the oracle source in 12/15 decisions. Within a routed source, however, logistic-MLP Kendall agreement is only $\tau = 0.1619$, compared with candidate-level $\tau = 0.3183$ and 67.92% sign agreement. The frozen positive-utility classifier attains mean utility $+0.00755$ with 26.7% harm, but the corresponding target means are $+0.02754$, -0.00356 , and -0.00132 . Its aggregate gain is therefore driven by one target. Two CPU runs reproduce all 1,200 utilities exactly in 83.55 and 82.56 seconds.

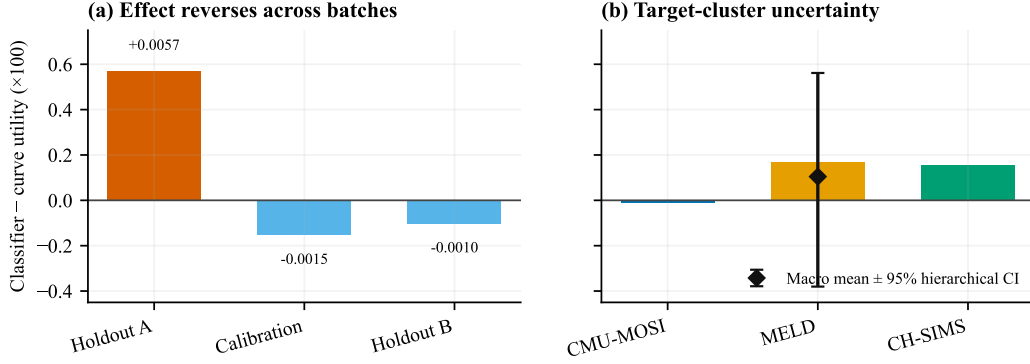


Figure 2: Selector gains depend on the repartition family. (a) The positive-utility classifier exceeds Curve-19 on the first sealed family, then reverses on the calibration and final families. (b) Target-level effects and the hierarchical-bootstrap interval, which includes zero.

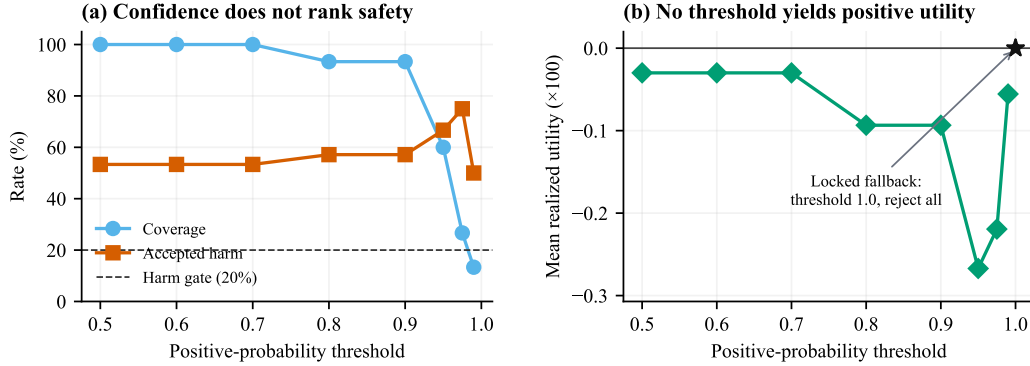


Figure 3: Classifier probability does not provide a useful safety threshold. Raising the positive-class threshold reduces coverage without controlling accepted harm or producing positive utility. The locked fallback rejects every candidate.

Learned fusion also disrupts the source-level result. Although the gated learner uses sample-dependent modality weights and attains nondegenerate anchor scores, the fixed distance route matches its oracle source in only 8/15 decisions. Its source-stratified agreement with logistic utility is $\tau = 0.1452$, with 63.75% sign agreement. The frozen positive-utility classifier and Curve-19 select mean utilities of -0.00027 and -0.00413 ; their harm rates are 53.3% and 73.3%. Two runs reproduce the same 1,200-row payload hash in 94.47 and 94.97 seconds. Both encoders remain frozen, so this panel tests learned fusion rather than end-to-end representation learning.

5.5 A fourth corpus does not rescue text-only bucket ranking

An auxiliary text-only experiment adds CMU-MOSEI (Zadeh et al., 2018) as a fourth independent target and source. This panel separates the small number of targets from the availability of aligned audio and contains 96 target-candidate pairs, or 480 five-seed utility evaluations. The full text signature reaches target-level Kendall $\tau = 0.1486$, with a 95% target-cluster interval of $[-0.1884, 0.4855]$ (Table 5). Text-centroid distance has a higher point estimate, $\tau = 0.2572$, whereas the full signature is negatively associated with utility within source, $\tau = -0.1369$. This experiment is not multimodal and cannot strengthen the main scope, but adding one target does not resolve the within-source ranking problem.

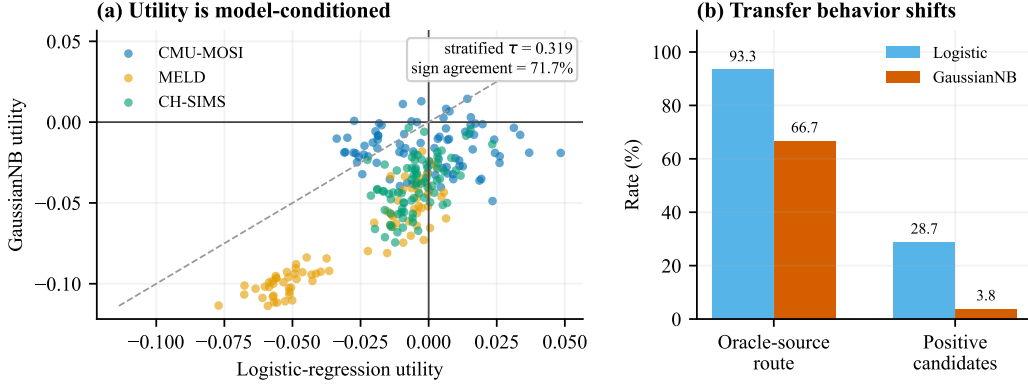


Figure 4: The downstream learner changes utility and routing. (a) Candidate utilities retain moderate rank agreement but shift downward under GaussianNB. (b) Oracle-source routing and the proportion of beneficial candidates both decline on the matched panel.

Table 3: Frozen selector decisions across downstream learners. Logistic results pool three sealed partition families (45 decisions); GaussianNB, the independent-head neural MLP, and learned gated fusion use the matched final family (15 decisions each). Utility is validation macro-F1 change. Route is oracle-source accuracy and is shared by methods within a model. Expected random and size-only results ($\dagger$) average ties analytically and are post-hoc descriptive baselines.

Model	Bucket rule	Utility $\uparrow$	Positive % $\uparrow$	Harm % $\downarrow$	Regret $\downarrow$	Route % $\uparrow$
Logistic	Positive-utility classifier	+0.00178	51.1	48.9	0.01727	97.8
Logistic	Curve-19	+0.00073	40.0	60.0	0.01832	97.8
Logistic	Uniform random $\dagger$	+0.00077	43.1	56.9	0.01810	97.8
Logistic	Size-only $\dagger$	-0.00050	40.8	59.2	0.01937	97.8
GaussianNB	Positive-utility classifier	-0.03734	0.0	100.0	0.02773	66.7
GaussianNB	Curve-19	-0.03744	0.0	100.0	0.02783	66.7
Neural MLP	Positive-utility classifier	+0.00755	73.3	26.7	0.01405	80.0
Neural MLP	Curve-19	+0.00486	66.7	33.3	0.01674	80.0
Gated fusion	Positive-utility classifier	-0.00027	46.7	53.3	0.02075	53.3
Gated fusion	Curve-19	-0.00413	26.7	73.3	0.02460	53.3

6 Discussion and Limitations

The main pattern is a hierarchy of predictability. Domain proximity carries enough signal to route between corpora under the logistic recipe, but it does not order buckets within the chosen source. Classification confidence does not repair this gap: thresholding reduces coverage without controlling harm. More fundamentally, the GaussianNB, MLP, and gated-fusion panels alter the prevalence and ranking of beneficial candidates. A data score detached from the downstream learner is thus an incomplete account of utility in this setting.

Several limitations bound that conclusion. The multimodal study has three independent target corpora, all drawn from sentiment or emotion tasks. Every panel uses frozen representations. The MLP broadens the architecture check, and gated fusion learns modality weights, but neither substitutes for end-to-end encoder fine-tuning. Utility is validation macro-F1 under one augmentation recipe; changing the source weight, metric, fusion rule, or training budget defines a new quantity. Finally, the natural repartitions reuse source examples across salts. They diversify decisions but do not create additional independent corpora.

Table 4: Matched cross-model utility diagnostics over 240 candidates, using logistic regression as reference. Candidate-level agreement overstates the within-stratum stability relevant to bucket selection.

Diagnostic	GaussianNB	Neural MLP	Gated fusion
Candidate Kendall τ	0.4222	0.3183	0.3503
Source-stratified Kendall τ	0.3190	0.1619	0.1452
Utility-sign agreement	71.67%	67.92%	63.75%
Positive candidates	3.75%	51.67%	45.00%
Positive candidates, logistic		28.75%	

Table 5: Auxiliary four-corpus text-only leave-one-target-out panel, adding CMU-MOSEI as a fourth independent target. Brackets are exploratory 95% target-cluster bootstrap intervals over only four targets. Lower regret is better.

Rule	Target Kendall τ	Top-1 regret	Within-source τ
Size-only	-0.3233 [-0.5695, -0.0764]	0.03573 [0.02466, 0.04681]	-0.0952
Length-only	+0.1449 [-0.0797, +0.3043]	0.02255 [0.01098, 0.03766]	+0.0536
Text-centroid-only	+0.2572 [+0.0362, +0.4783]	0.02183 [0.00801, 0.03564]	-0.1369
Full text signature	+0.1486 [-0.1884, +0.4855]	0.00583 [0.00180, 0.01022]	-0.1369

7 Conclusion

Prospective multimodal data selection must specify both the downstream learner and the granularity of the decision. In this study, corpus routing is reliable for the logistic recipe, while within-source selection and calibrated abstention are not. The matched learners further show that utility prevalence, routes, and bucket rankings need not transfer across models. Claims of general-purpose selection therefore require learner-conditioned benchmarks, uncertainty over independent targets, and safety results that report coverage rather than abstention alone.

AI use statement

Generative AI assisted with hypothesis and methodology refinement, literature discovery, translation, code and figure drafting, data-artifact formatting, result interpretation, and manuscript editing. It was not used to create training or evaluation examples, ground-truth labels, mathematical claims, or proofs. The authors checked citations against primary records, executed and tested generated code locally, and verified reported numbers against versioned machine-readable artifacts. The authors reviewed all assisted work and take responsibility for the final text, claims, code, and artifacts.

Ethics statement

This work conducts secondary analysis of existing multimodal sentiment and emotion corpora and recruits no new participants. Raw speech, video, and transcripts remain subject to the access conditions and licenses of the original corpora; supplementary materials will distribute only code and derived artifacts permitted by those terms. Affective labels and representations can encode demographic, linguistic, cultural, and annotator biases. Dataset-utility rankings should therefore not be interpreted as judgments about the quality or worth of speakers, nor used to justify high-stakes affect inference. Our conclusions are limited to the audited benchmark tasks, representations, models, metrics, and training recipes.

Reproducibility statement

Section 3 specifies the problem and information boundary. Section 4 records the datasets, representations, signatures, partitions, seeds, downstream recipes, decision rules, uncertainty analysis, and compute accounting. Figures and tables are generated from versioned machine-readable artifacts. Subject to corpus licenses, the anonymized supplement will in-

clude code for data audits, utility panels, selectors, statistical analyses, tables, and figures, together with artifact hashes.

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